

ECM3446

UNIVERSITY OF EXETER

**FACULTY OF ENVIRONMENT, SCIENCE
AND ECONOMY**

COMPUTER SCIENCE

Examination, May 2023

High Performance Computing

Module Leader: Man Luo

Duration: TWO HOURS

No Word Count Specified

Answer ALL questions.

The marks for this module are calculated from 60% of the percentage mark for this paper plus 40% of the percentage mark for associated coursework.

Candidates are permitted to use approved portable electronic calculators in this examination.

This is a CLOSED BOOK examination.

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Question 1

(a) The Top500 list ranks the fastest supercomputers in the world and has been running since 1993.

(i) What is the performance benchmark used for the Top500 list?

(2 marks)

(ii) What type of workload does the performance benchmark for Top500 list use?

(2 marks)

(iii) "Given current trends soon all High Performance Computing systems are going to be commodity clusters." Argue for **OR** against this position.

(4 marks)

(b) Effective load balancing can significantly improve the performance of a parallel program.

(i) Define what is meant by parallel efficiency?

(2 marks)

(ii) Explain what is meant by "load balancing".

(4 marks)

(iii) How can load balancing be improved when parallelising a loop with OpenMP?

(2 marks)

(c) A C program, which is parallelised using MPI, contains the following code fragment

```
if (myRank == 0)
    printf("The result is %d\n", globalSum);
else
    /*Wait for the result to be printed out */
    MPI_Barrier (MPI_COMM_WORLD);

}
```

You know that the variable `myRank` is an integer variable which holds the process's rank within the `MPI_COMM_WORLD` communicator.

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(i) What happens when the program reaches this section of code and why?

(6 marks)

(ii) What type of bug is this?

(2 marks)

(d) Matrix operations are often used in high performance computing to solve systems of linear equations.

(i) What is the difference between a sparse matrix and a dense matrix?

(2 marks)

(ii) Describe the compressed sparse row format for storing sparse matrices.

(4 marks)

(iii) How does the number of bytes of data transferred from memory scale with N for dense matrix multiplication and why?

(2 marks)

(e) You are given the task of helping a researcher speed up their program and you suggest parallelising the program. The researcher says: “Parallel programming is too difficult. I’ve heard that Moores Law is still going. Why can’t I get a much faster processor to make my serial program run quicker?”

How would you explain to them that they have mis-understood Moores Law and why they can’t get a faster single processor core?

(8 marks)

(Total 40 marks)

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Question 2

- (a) The following C program calculates the mean and variance of an array of values. It also finds the maximum value in the array.

```
#include <stdio.h>

#define N 5

int main(){

    float x[N]={1.0, 4.0, 6.0, 9.0, 3.0};
    float sum, sum2;
    float mean, sigma2, max_val;
    int i;

    sum      =0.0;
    sum2     =0.0;
    max_val  =-1.0e37;

    for (i=0; i<N; i++){
        sum  +=x[i];
        sum2 +=(x[i] * x[i]);
        if (x[i] > max_val) {max_val = x[i];}
    }

    mean  = sum / ( (float) N);
    sigma2 = (sum2 / ( (float) N) ) - (mean * mean);

    printf("Mean = %g\n", mean);
    printf("Sigma^2= %g\n", sigma2);
    printf("Max= %g\n", max_val);

    return 0;
}
```

Show how to parallelise the `for` loop using OpenMP.

(10 marks)

- (b) Compare the OpenMP and MPI methods for writing parallel programs. Discuss the strengths and weaknesses of each method.

(10 marks)

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- (c) Show how the following sparse matrix would be represented in compressed sparse row format with rows and columns numbered starting from one.

(10 marks)

$$\begin{pmatrix} 6 & 0 & 7 & 0 \\ 0 & 2 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(Total 30 marks)

Question 3

- (a) You have been given the task of parallelising a program and need to achieve a strong scaling speed-up of at least 100.

- (i) What is the smallest parallel fraction which permits a speed-up of 100?

(3 marks)

- (ii) You parallelise the program and achieve a parallel fraction of $p = 0.999$. How many processors are required to obtain a speed up of 100 assuming idealised strong scaling?

(3 marks)

- (iii) If the total parallel overhead is equivalent to 0.5% of the serial run time, what is the smallest number of processors required to achieve a speed up of at least 100?

(4 marks)

- (b) You are given the task of planning an upgrade to an HPC cluster which involves replacing all the compute nodes. The total number of compute nodes will not change and there will be no other changes to the hardware. You need to choose between two compute node specifications:

- Compute node type 1: number of sockets=2, number of cores per socket=64, processor clock frequency=2.2 GHz, floating point operations per cycle =4
- Compute node type 2: number of sockets=2, number of cores per socket=28, processor clock frequency=2.8 GHz, floating point operations per cycle =16

The cost of the compute nodes is the same so you need to make a choice based on performance.

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- (i) Which type of compute node has the higher theoretical peak performance? Show your working.

(5 marks)

- (ii) You know that the application which will run on the system has not been optimised to make effective use of the instruction set in modern processors. This means that the application is not able to use vector instructions and fused-multiply-add instructions. Explain whether this would influence your choice of compute node and why.

(3 marks)

- (iii) You know that the application which will run on the system makes extensive use of sparse matrix operations. What extra piece of information would you need to know in order to assess the suitability of the compute nodes for running this application and why is this information important?

(2 marks)

- (c) Consider a two-dimensional partial differential equation (PDE) solver which solves a PDE with one unknown variable u . To solve the PDE it is necessary to calculate values for the second derivative

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

This expression is evaluated using the finite difference stencil

$$\text{del2u} = (u(i+1, j) - 2*u(i, j) + u(i-1, j)) / (dx*dx) + (u(i, j+1) - 2*u(i, j) + u(i, j-1)) / (dy*dy)$$

The computation domain for the whole calculation comprises 8192×8192 grid points. In addition to these grid points each sub-domain stores halo values which are copies of the data on neighbouring processors.

- (i) The calculation is domain decomposed in an 8×8 configuration such that there are 64 square sub-domains of equal size. How many values need to be exchanged when one halo is sent to one neighbouring process?

(2 marks)

- (ii) How many bytes need to be sent if the values are represented using IEEE754 double precision?

(2 marks)

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- (iii) What is the transmission time for sending a single halo over a network with a latency of $1\mu\text{s}$ (10^{-6}s) and a bandwidth of 10.5 GB/s (10.5 gigabytes per second)? Show your working.

(3 marks)

- (iv) The domain decomposition is modified so that the domain is decomposed in a 16×16 configuration such that there are 256 square sub-domains of equal size. What is the transmission time for one halo with the modified domain decomposition?

(3 marks)

(Total 30 marks)